

MMC simple, average and switching models simulation

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Chapter 1

MMC structure

This document explains the simulations of three different MMC models in MATLAB-Simulink: the simple, average and switched models.

1.1 Converter structure

The structure of the converter is as follows in [Figure 1.1](#)

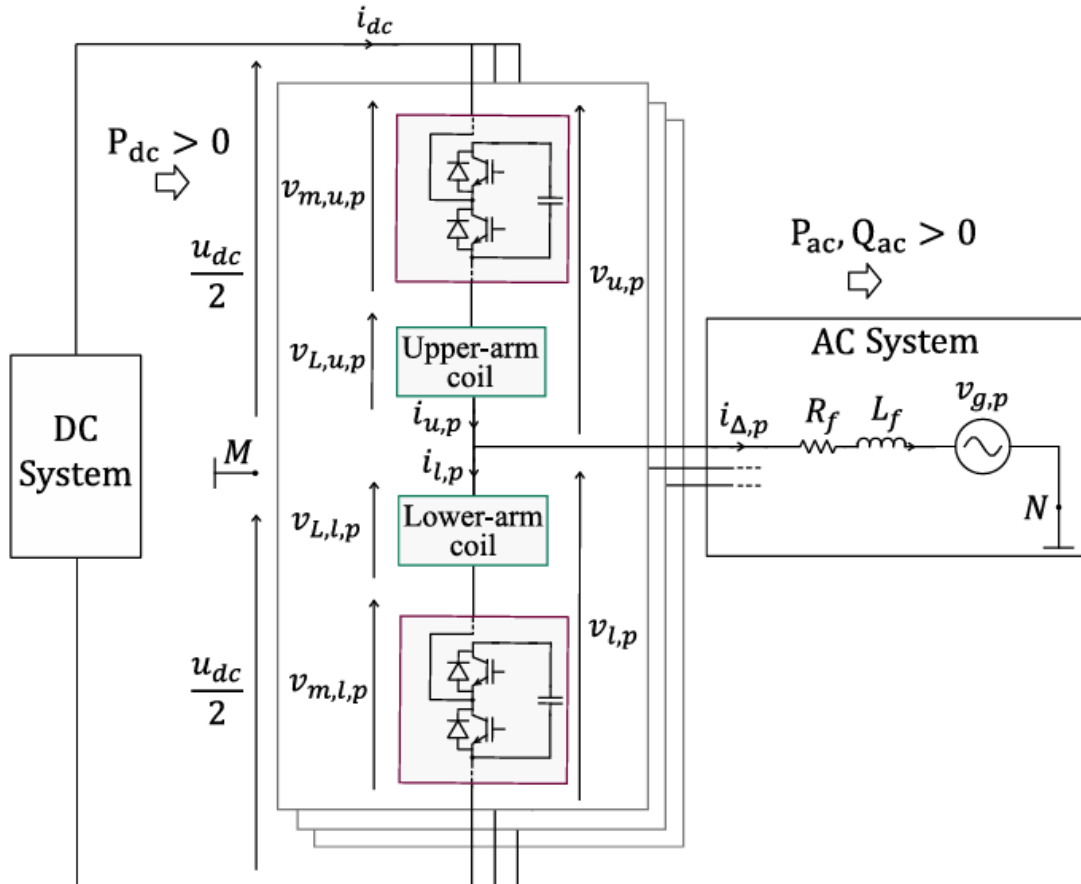


Figure 1.1: MMC circuit + AC system

The indices u and l refer to "upper" and "lower," respectively. The index p corresponds to phase p , and the index i corresponds to the i -th module. It is assumed that the converter has N submodules (SM) per arm.

1.2 Modules

For these MMC simulations, we consider that each SM is a half-bridge consisting of two switches and one capacitor.

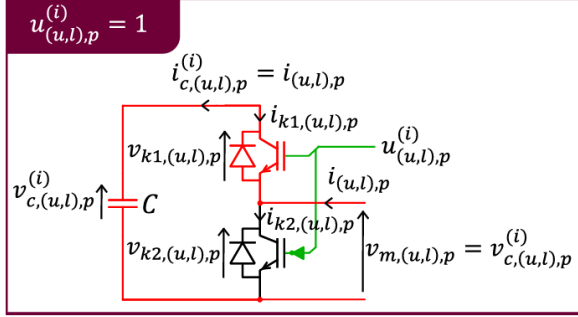


Figure 1.2: Inserted SM

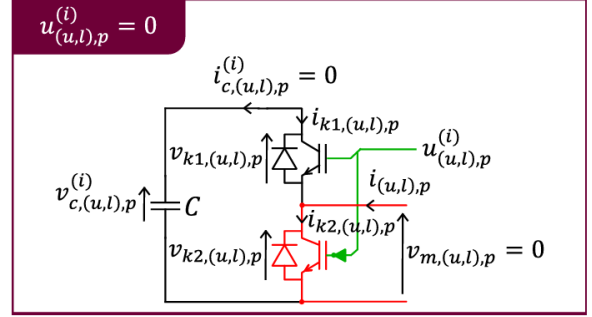


Figure 1.3: Bypassed SM

The SMs have two distinct states. [Figure 1.2](#) when it is inserted into the arm. While in this state it adds its voltage $v_{m,(u,l),p}^{(i)}$ to the arm voltage $v_{m,(u,l),p}$. [Figure 1.3](#) when it is bypassed, such that the SM does not add to the arm voltage $v_{m,(u,l),p}$.

The SM state is controlled by their switching function $u_{(u,l),p}^{(i)}$, that gives the commutation pulses for $K1$ and $K2$.

1.3 MMC design parameters used on simulations

The parameters defining the application ratings and MMC design for the simulations described in this document are detailed in [Table 1.1](#).

Table 1.1: Simulation Parameters

Parameter	Symbol	Value
MMC Parameters		
Arm resistance	R_{arm}	$0.01 \, \Omega$
Arm inductance	L_{arm}	$1 \, \text{mH}$
Equivalent capacitor capacitance	C_{eq}	$3 \, \text{mF}$
Number of modules per arm	N	6
External Parameters		
Electrical frequency	f_0	$50 \, \text{Hz}$
Motor inductance	L_{ac}	$100 \, \mu\text{H}$
Motor resistance	R_{ac}	$1 \, \Omega$
DC cable resistance	R	$10^{-6} \, \Omega$
DC voltage	V_{DC}	$800 \, \text{V}$
Simple Model Parameters		
Modulation index	m	1
SM state of charge	a	1
AVM and SW Model Parameters		
	$m_{\Delta,0}$	0
	$m_{\Sigma,0}$	1
	M_{Δ}	1
	Φ_{Δ}	0
	M_{Σ}	0
	Φ_{Σ}	0

Chapter 2

MMC simple model

2.1 Model on simulink

The MMC simple model of Figure 2.1 is the simplest model of the MMC since it does not represents any SM capacitor dynamics, having an ideal operation of the converter.

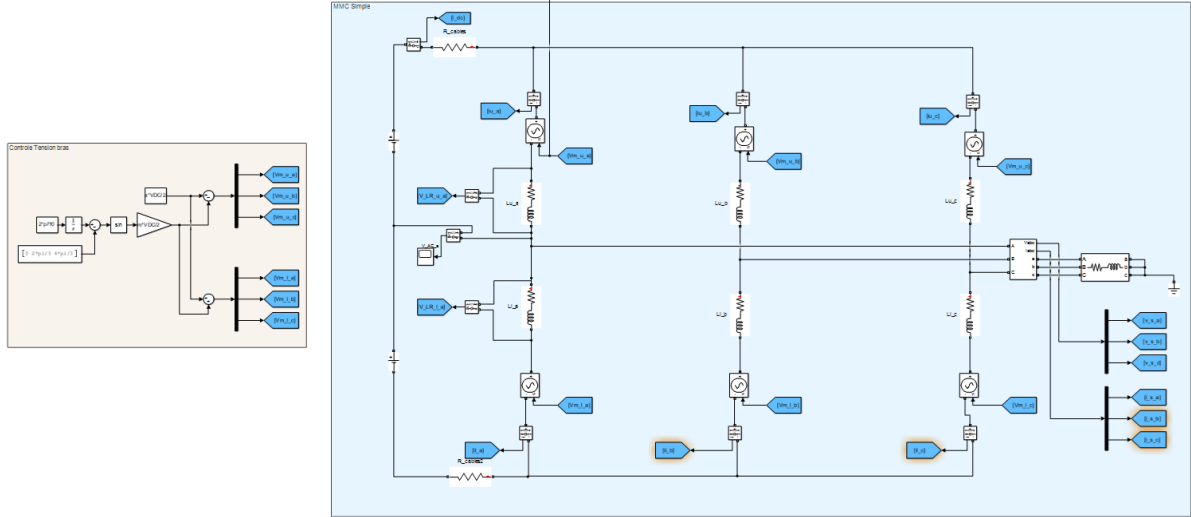


Figure 2.1: MMC simple model

It uses one equivalent controlled voltage source to represent the arm voltage $v_{m,(u,l),p}$. The controlled voltage sources for each arm have their output directly given by the modulation reference defined on the left.

Figure 2.2 shows the modulation reference calculation in detail. It can be translated by (2.1), where m is the modulation index and the additional parameter a corresponds to the state of charge of the SM.

$$v_{m,u,p}(t) = a \frac{V_{DC}}{2} - m \frac{V_{DC}}{2} \sin(\omega t + \Phi_p) \quad (2.1)$$

By performing the KVL on the upper arm loop, we obtain that the voltage output $v_{AC,p}$, measured between the phase p and node M, is:

$$v_{AC,p} = \frac{V_{DC}}{2} - v_{m,u,p} - v_{L,u,p} \quad (2.2)$$

Therefore, by applying the modulation equation, the output voltage is:

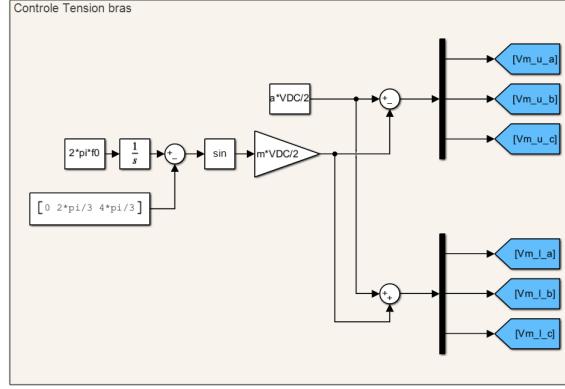


Figure 2.2: Modulation for simple model

$$v_{AC,p} = (1 - a) \frac{V_{DC}}{2} + m \frac{V_{DC}}{2} \sin(\omega t + \Phi_p) - v_{L,u,p} \quad (2.3)$$

From this equation, we can deduce that there is a voltage drop on the arm inductance that will also compensate for the DC level $(1 - a) \frac{V_{DC}}{2}$ if the SMs state of charge is not ideal, which will be demonstrated through simulations results.

2.2 Simulations

2.2.1 Original specifications

First, a simulation with the original specifications is performed. Using (2.3), the AC output voltage should have the amplitude of $V = m \frac{V_{DC}}{2} = 400V$ neglecting inductor losses. Figure 2.3 shows that the amplitude measured for phase a , $V_{AC,a} = 391,4V$, is close to the expected value.

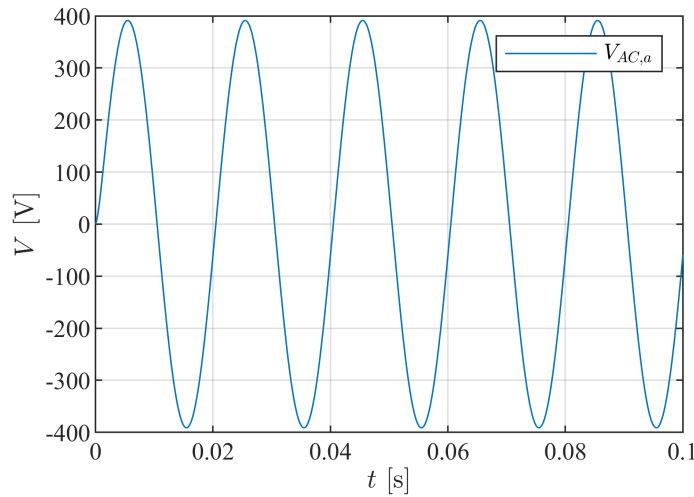


Figure 2.3: $v_{AC,a}$, AC output voltage for phase a

In terms of circulating current, in Figure 2.4 we can see that $i_{\Sigma,a} = 0$, which is coherent with the state of charge parameter $a = 1$. The dc current is also 0.

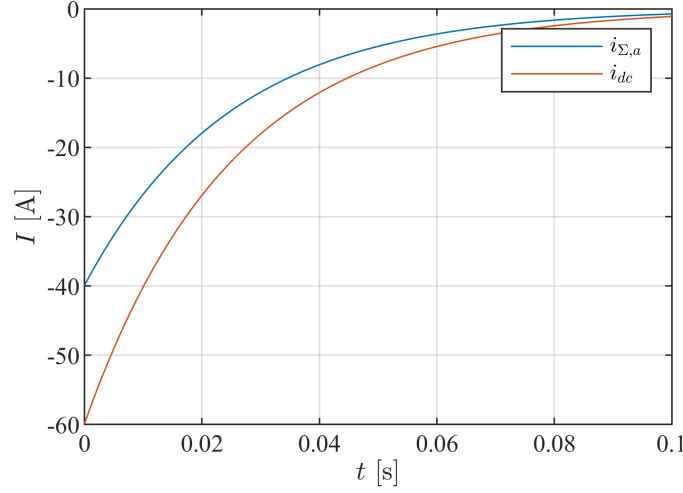


Figure 2.4: Circulating and DC currents

Comment : The dc current being 0 is a little bit strange for me because it would mean that no power is transferred from the DC side to the AC side, but if we sum all the currents in the arm we also have $i_{dc} = i_{u,a} + i_{u,b} + i_{u,c} = 0$ since they don't have a dc level.

We can also observe in Figure 2.5 the voltage on the arm coils $v_{L,u,a}$ meaning that some power is dissipated there, which explains the difference between measured $V_{AC,a}$ amplitude and the expected value of 400 V.

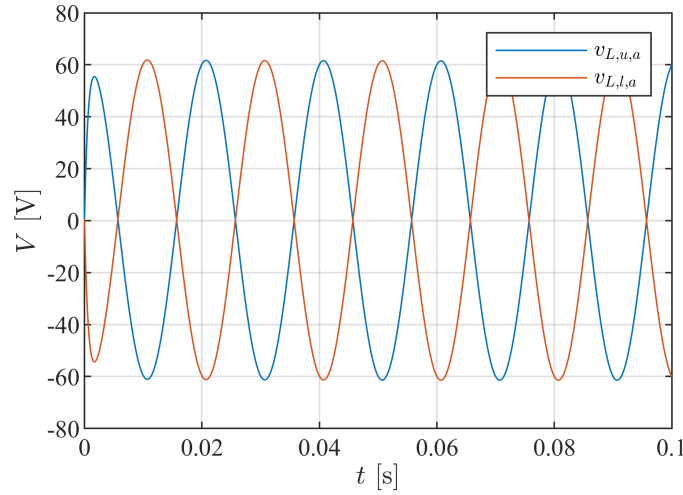


Figure 2.5: Arm coil voltage

The voltage, current and power on the AC side load in Figure 2.6 shows sinusoidal ideal outputs, which is consistent with the simple model usage that does not include harmonics generation from capacitor dynamics. The power output on phase a is 76,5 kW, which is equivalent to a power output 229,5 kW for the MMC.

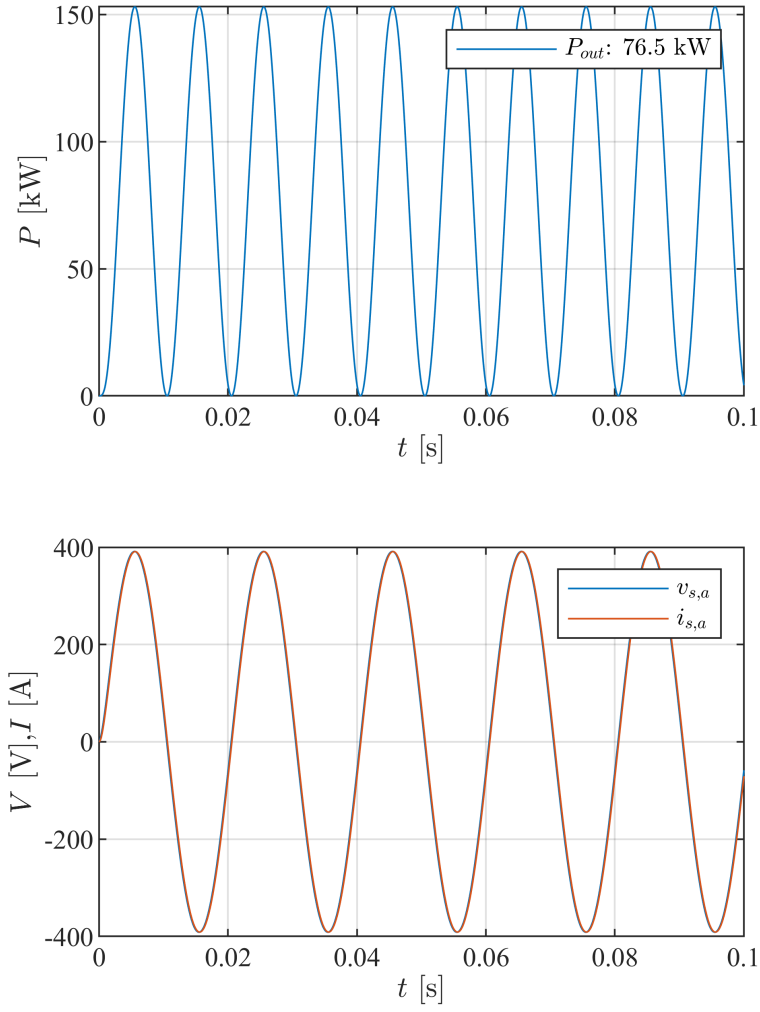


Figure 2.6: AC load outputs

2.2.2 Modifying m

Figure 2.7 and Figure 2.8 shows simulation results with $m = 0, 5$ compared to simulation with $m = 1$ using the original parameters. In Figure 2.7 we can see that $v_{s,a}$ amplitude is reduced to a value near $V = m \frac{V_{DC}}{2} = 200 \text{ V}$ as expected. Also, the power on the 3-phase load is reduced to 57,3 kW.

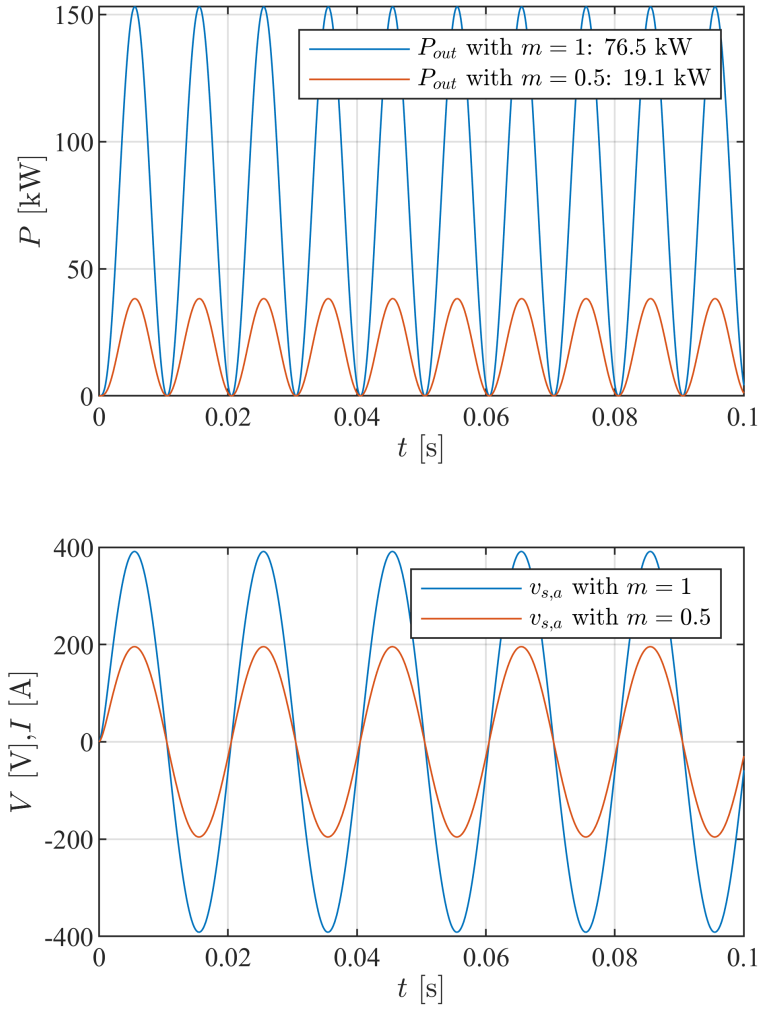


Figure 2.7: AC load outputs with $m = 0, 5$

In [Figure 2.8](#) we observe that the voltage drop on the arm coils is reduced since less current passes through the arms, such that the losses on the arm coils is also reduced.

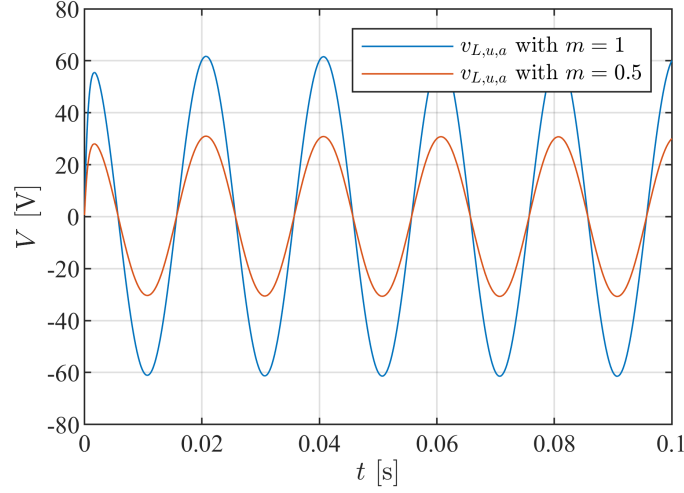


Figure 2.8: Arm coil voltage with $m = 0,5$

2.2.3 Modifying a

Figure 2.9 and Figure 2.10 shows simulation results with $a = 0,5$ compared to simulation with $a = 1$ using the original parameters. In Figure 2.9 we can see that $i_{\Sigma,a} \neq 0$ and a DC circulating current appears. $i_{\Sigma,a}$, it has only the DC component since no capacitor is considered in this model. We also observe that $i_{dc} \neq 0$.

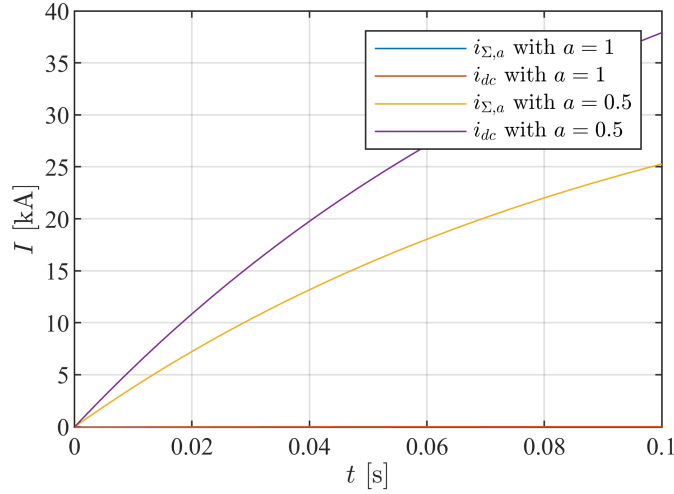


Figure 2.9: AC load outputs with $a = 0,5$

This happens because $a = 0,5$ means that the SM are not charged to its mean value, such that there is a DC voltage on the arm's coil. The DC level voltage on the arm coils can be deduced from (2.3) :

$$\langle v_{L,u,p} \rangle = (1 - a) \frac{V_{DC}}{2} = \frac{800}{4} = 200V \quad (2.4)$$

$$\langle v_{L,l,p} \rangle = (a - 1) \frac{V_{DC}}{2} = -\frac{800}{4} = -200V \quad (2.5)$$

In Figure 2.10 we observe that $v_{L,u,a}$ DC voltage level is 200 V, as expected.

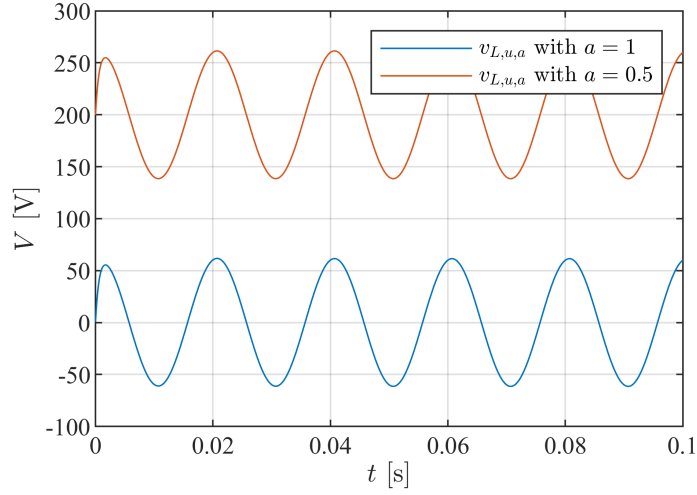


Figure 2.10: Arm coil voltage with $a = 0,5$

2.2.4 Modifying R_{load} and L_{load}

By making the load more inductive with $L_{load} = 1$ mH in [Figure 2.11](#), a phase shift between the output voltage and current occurs. Because of that, the active output power on the load is reduced.

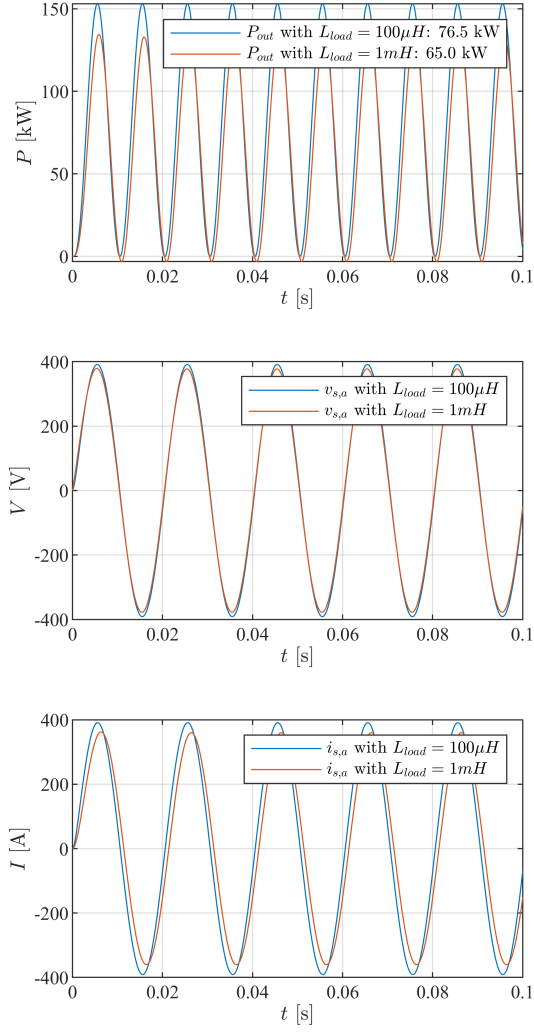


Figure 2.11: AC load outputs increasing L_{load}

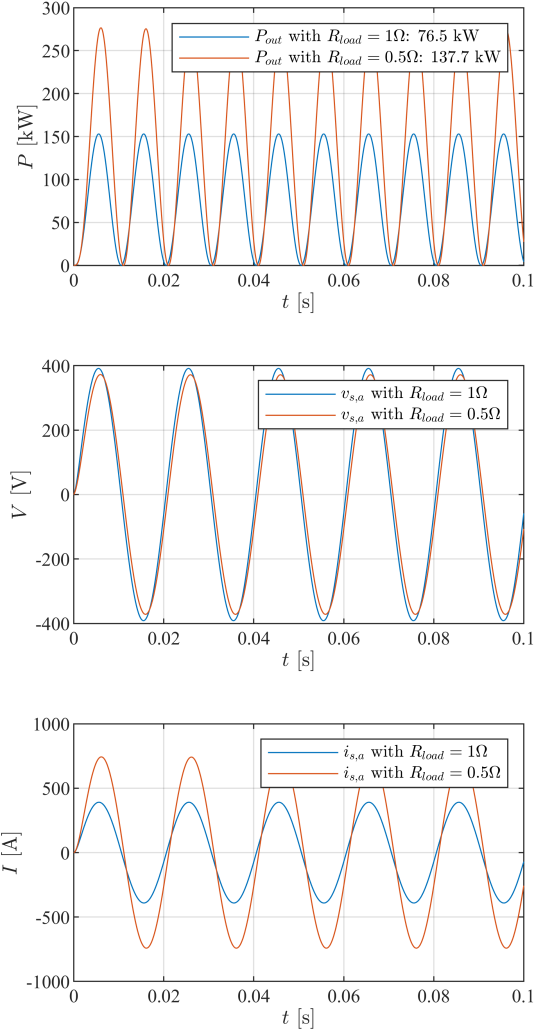


Figure 2.12: AC load outputs reducing R_{load}

In Figure 2.12, by reducing R_{load} to $0,5 \Omega$, we can notice that the load current increases while the output voltage is maintained in practically the same level since it is controlled by the modulation inputs. Consequently, the output power increases.

Because of the current increase, a higher voltage drop occurred across the arm coils as shown in Figure 2.13. Consequently, the output voltage amplitude is reduced to 371,7 V instead of 391,4 V.

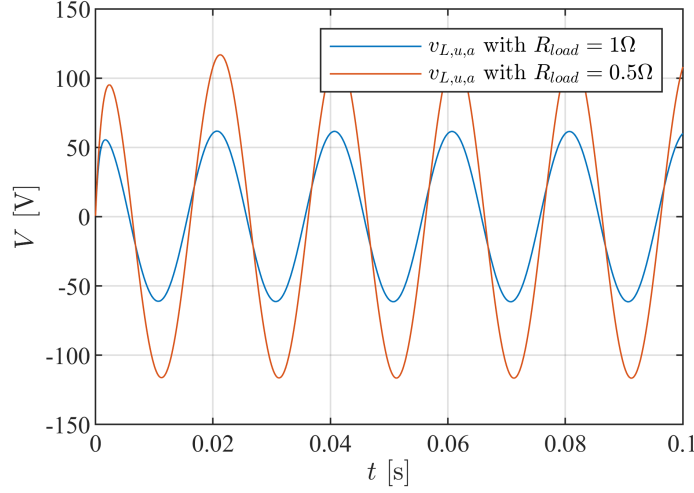


Figure 2.13: AC load outputs with $R_{load} = 0,5\Omega$

2.3 Summary

The simple model is the simplest model of the MMC and represents the MMC ideal operation. The arm voltages are approximated by a sinusoidal signal. Therefore it does not include switching effects neither circulating currents harmonics.

From simulation results, we conclude that the phase output voltage amplitude is related to the modulation index parameter m and the DC component of the circulating current is related to the average level of charge of the capacitors a parameter, as demonstrated in the “simple” model simulation. We can also observe that, by increasing L_{load} inductance, the phase-shift between phase a voltage and current outputs causes a reduction in the load active power. By reducing R_{load} , the current output becomes higher which implies more power output but also an increased voltage drop on the arm coils.

Chapter 3

MMC average model

3.1 Model on simulink

The MMC average model schematic of [Figure 3.1](#) shows that the average model has a step up from the MMC simple model since it represents the SMs capacitor dynamics.

The SMs capacitor dynamics are represented by an equivalent capacitor within each arm. This approximation is possible by assuming that a great number of SMs is used and that the capacitor voltage of each SM within one arm is balanced.

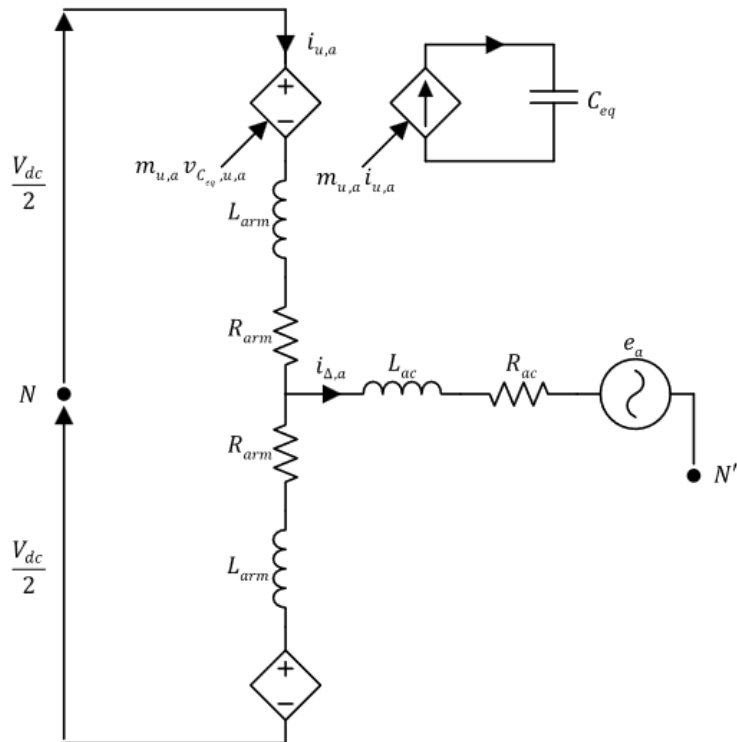


Figure 3.1: MMC average model schematic

The MMC average model simulation built in MATLAB-Simulink shown in [Figure 3.2](#) is composed of several different blocks.

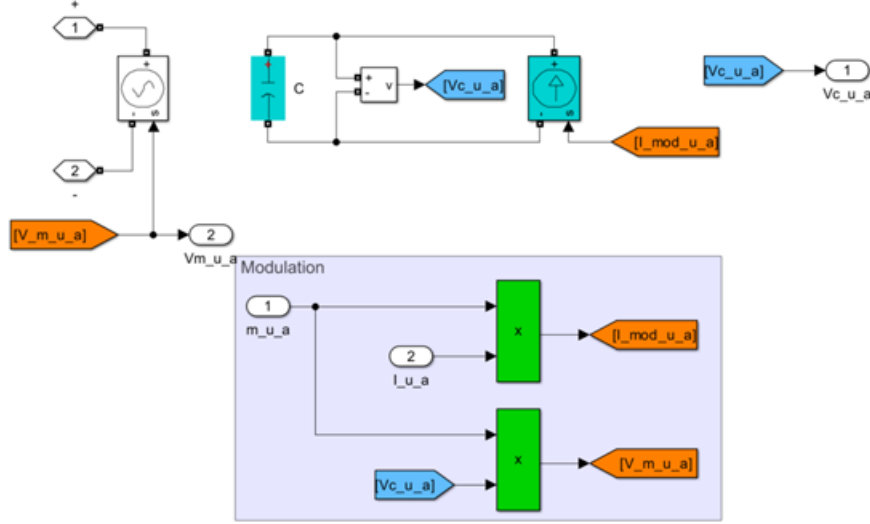


Figure 3.4: Average model arm subsystem on Simulink

The voltage source output is proportional to the equivalent capacitor voltage and controlled by the modulation function m , such that

$$v_{m,u,a} = m_{u,a} \cdot v_{c,u,a} \quad (3.1)$$

As previously explained, the arm dynamics is represented by a single equivalent capacitor. The capacitor is charged or discharged according to the arm current and the modulation function, such that

$$i_{m,u,a} = m_{u,a} \cdot i_{u,a} \quad (3.2)$$

To implement this dynamics, a current source is used in series with the equivalent capacitor. The equivalent capacitor voltage is considered to be equal to N times an individual module capacitor voltage.

3.1.2 $\Delta\Sigma$ transform

To go from $ul \rightarrow \Delta\Sigma$ variables, we use the $\Delta\Sigma$ variable transform defined as follow:

$$\begin{pmatrix} v_{m,\Delta,p} \\ v_{m,\Sigma,p} \\ i_{\Delta,p} \\ i_{\Sigma,p} \\ v_{c,\Delta,p} \\ v_{c,\Sigma,p} \\ m_{\Delta,p} \\ m_{\Sigma,p} \end{pmatrix} \triangleq \begin{pmatrix} 1/2 & -1/2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & -1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} v_{m,l,p} \\ v_{m,u,p} \\ i_{l,p} \\ i_{u,p} \\ v_{c,l,p} \\ v_{c,u,p} \\ m_{l,p} \\ m_{u,p} \end{pmatrix} \quad (3.3)$$

Its inverse transform is used to go from $\Delta\Sigma \rightarrow ul$ variables.

This transformation was developed in [3] because working with traditional MMC modeling that uses ul, abc references for MMC control is not easy. This transformation is used to allow steady-state time-invariant (SSTI) state space representation of the MMC by separating the state and control variables into two groups: those mainly oscillating at -2ω

(noted with Σ) and those oscillating at $+\omega$ (noted with Δ). To achieve the SSTI representation, the Park transform is applied to the Δ and Σ quantities at the corresponding frequency, $+\omega$ and -2ω respectively.

I used here the notations developed afterwards from [1] and [2].

3.1.3 Park transform

To go from $abc \rightarrow dq0$ frame, we use the Park transform defined as follows:

$$\begin{pmatrix} x_d \\ x_q \\ x_0 \end{pmatrix} = \frac{2}{3} \begin{pmatrix} \cos(\theta) & \cos\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta - \frac{4\pi}{3}\right) \\ -\sin(\theta) & -\sin\left(\theta - \frac{2\pi}{3}\right) & -\sin\left(\theta - \frac{4\pi}{3}\right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x_a \\ x_b \\ x_c \end{pmatrix} \quad (3.4)$$

Its inverse transform is used to go from $dq0 \rightarrow abc$ frame. As explained before, the Park transform is applied to the Δ and Σ quantities at the corresponding frequency, $+\omega$ and -2ω respectively.

3.1.4 Modulation

The default modulation given parameters is:

$$\begin{cases} m_{\Delta,d} = 0 \\ m_{\Delta,q} = -1 \\ m_{\Delta,0} = 0 \\ m_{\Sigma,d} = 0 \\ m_{\Sigma,q} = 0 \\ m_{\Sigma,0} = 1 \end{cases} \quad (3.5)$$

This parameter combination produces a fundamental frequency modulation with amplitude 1 p.u. and DC component = 0.5 p.u., which is confirmed by the simulated $m_{u,abc}$ and $m_{l,abc}$ in Figure 3.5. Observe that $m_{u,abc}$ and $m_{l,abc}$ modulations are opposed to each other and centered at 0.5 p.u.

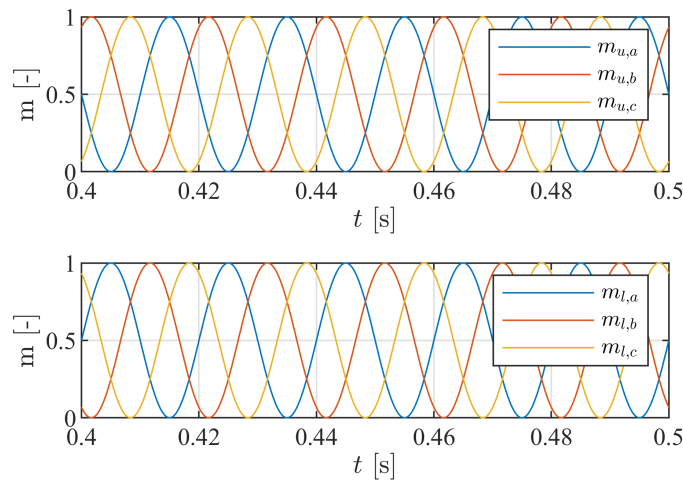


Figure 3.5: Modulation in *ulabc* frame simulation results

This can be derived mathematically from the MMC's equations as follows. From (??), using the inverse transform the $m_{u,abc}$ and $m_{l,abc}$ can be expressed in terms of Δ and Σ quantities:

$$m_{u,abc} = \frac{m_{\Sigma,abc}}{2} - \frac{m_{\Delta,abc}}{2} \quad (3.6)$$

Using that $m_{\Delta,abc} = P^{-1}(\theta = \omega t) \cdot m_{\Delta,dq0}$ and $m_{\Sigma,abc} = P^{-1}(\theta = -2\omega t) \cdot m_{\Sigma,dq0}$ and simplifying the equations, we find that

$$m_{u,a} = \frac{m_{\Sigma,0}}{2} - \frac{m_{\Delta,0}}{2} - \frac{\cos(\omega t) m_{\Delta,d}}{2} + \frac{\cos(2\omega t) m_{\Sigma,d}}{2} + \frac{\sin(\omega t) m_{\Delta,q}}{2} + \frac{\sin(2\omega t) m_{\Sigma,q}}{2} \quad (3.7)$$

$$\begin{aligned} m_{u,b} = & \frac{m_{\Sigma,0}}{2} - \frac{m_{\Delta,0}}{2} + \frac{\cos(\omega t) m_{\Delta,d}}{4} - \frac{\cos(2\omega t) m_{\Sigma,d}}{4} - \frac{\sin(\omega t) m_{\Delta,q}}{4} \\ & - \frac{\sin(2\omega t) m_{\Sigma,q}}{4} - \frac{\sqrt{3} \cos(\omega t) m_{\Delta,q}}{4} - \frac{\sqrt{3} \sin(\omega t) m_{\Delta,d}}{4} \\ & - \frac{\sqrt{3} \sin(2\omega t) m_{\Sigma,d}}{4} + \frac{\sqrt{3} m_{\Sigma,q} (2 \cos^2(\omega t) - 1)}{4} \end{aligned} \quad (3.8)$$

$$\begin{aligned} m_{u,c} = & \frac{m_{\Sigma,0}}{2} - \frac{m_{\Delta,0}}{2} + \frac{\cos(\omega t) m_{\Delta,d}}{4} - \frac{\cos(2\omega t) m_{\Sigma,d}}{4} \\ & - \frac{\sin(\omega t) m_{\Delta,q}}{4} - \frac{\sin(2\omega t) m_{\Sigma,q}}{4} + \frac{\sqrt{3} \cos(\omega t) m_{\Delta,q}}{4} \\ & - \frac{\sqrt{3} \cos(2\omega t) m_{\Sigma,q}}{4} + \frac{\sqrt{3} \sin(\omega t) m_{\Delta,d}}{4} + \frac{\sqrt{3} \cos(\omega t) \sin(\omega t) m_{\Sigma,d}}{2} \end{aligned} \quad (3.9)$$

Given the input parameters $m_{\Delta,q} = -1$, $m_{\Sigma,0} = 1$, $m_{\Delta,d} = m_{\Delta,0} = m_{\Sigma,d} = m_{\Sigma,q} = 0$, then, for phase a:

$$\begin{aligned} m_{u,a} &= \frac{m_{\Sigma,0}}{2} - \frac{m_{\Delta,0}}{2} - \frac{\cos(\omega t) m_{\Delta,d}}{2} + \frac{\cos(2\omega t) m_{\Sigma,d}}{2} + \frac{\sin(\omega t) m_{\Delta,q}}{2} + \frac{\sin(2\omega t) m_{\Sigma,q}}{2} \\ &= \frac{1}{2} - 0 - 0 + 0 + \frac{\sin(\omega t)(-1)}{2} + 0 \\ &= \frac{1}{2} - \frac{\sin(\omega t)}{2}. \end{aligned} \quad (3.10)$$

Performing the same calculations for the lower arm:

$$m_{l,a}(t) = \frac{1}{2} + \frac{\sin(\omega t)}{2} \quad (3.11)$$

These expressions corroborate with the obtained results for modulation in $ulabc$ frame in [Figure 3.5](#)

3.2 Simulations

3.2.1 Results in ul, abc frame

Performing a first simulation using the default parameters of [Table 1.1](#), the abc output voltages and currents shown in [Figure 3.6](#) are no longer sinusoidal and have a 2ω component.

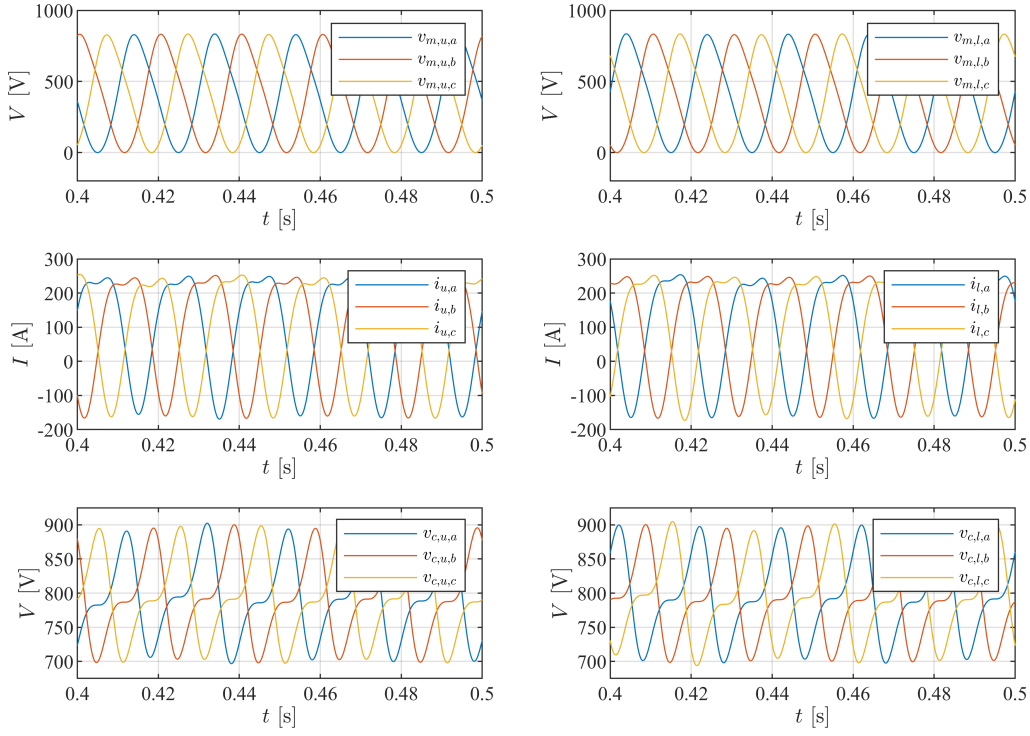


Figure 3.6: *ulabc* frame outputs simulation results

Analyzing each plot individually, the equivalent arm voltages $v_{m,u,abc}$ and $v_{m,l,abc}$ are not sinusoidal as expected by the introduction of the modulation with mainly fundamental frequency because of the 2ω harmonics introduced. The arm currents $i_{u,abc}$ and $i_{l,abc}$ are deformed which is also related to the harmonics introduced. The equivalent capacitor voltages $v_{c,u,abc}$ and $v_{c,l,abc}$ have a medium stable stage near its mean DC voltage level of 800 V. In terms of charging and discharging dynamics, we observe that the capacitor voltage increases when the arm current is positive, and the capacitor voltage decreases when the arm current is negative.

3.2.2 Results in $\Delta\Sigma, abc$ frame

The presence of the 2ω circulating current is even clearer when analyzing the $\Delta\Sigma$ simulation outputs shown in Figure 3.8. As described in [2], in the usual operation of the MMC, the Σ quantities have frequency of 2ω and a DC level. Also, the Δ quantities have only fundamental frequency.

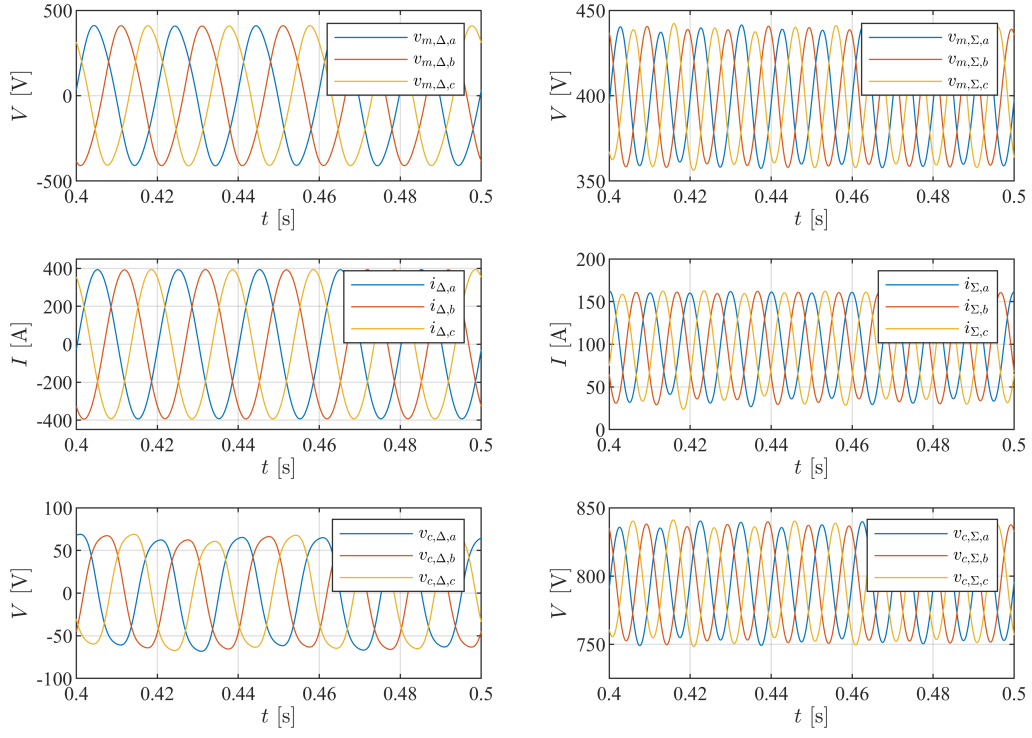


Figure 3.7: $\Delta\Sigma abc$ frame outputs simulation results

3.2.3 Results in $\Delta\Sigma, dq0$ frame

In [Figure 3.8](#) we verify that the results in $dq0$ frame are almost constant with very little oscillation, which is consistent with the use of Park's transform.

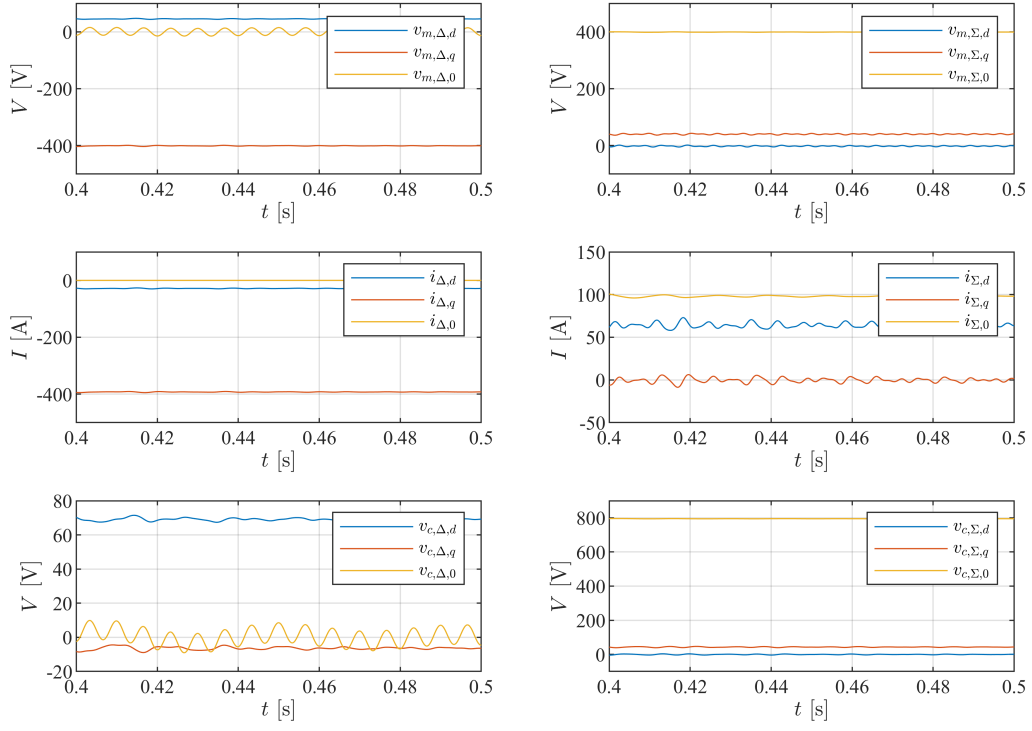


Figure 3.8: $\Delta\Sigma dq0$ frame outputs simulation results

3.2.4 Modifying L_{arm}

Reducing L_{arm} from 1 mH to 100 μH , we observe in the simulations results of [Figure 3.9](#) that the filtering capability is reduced.

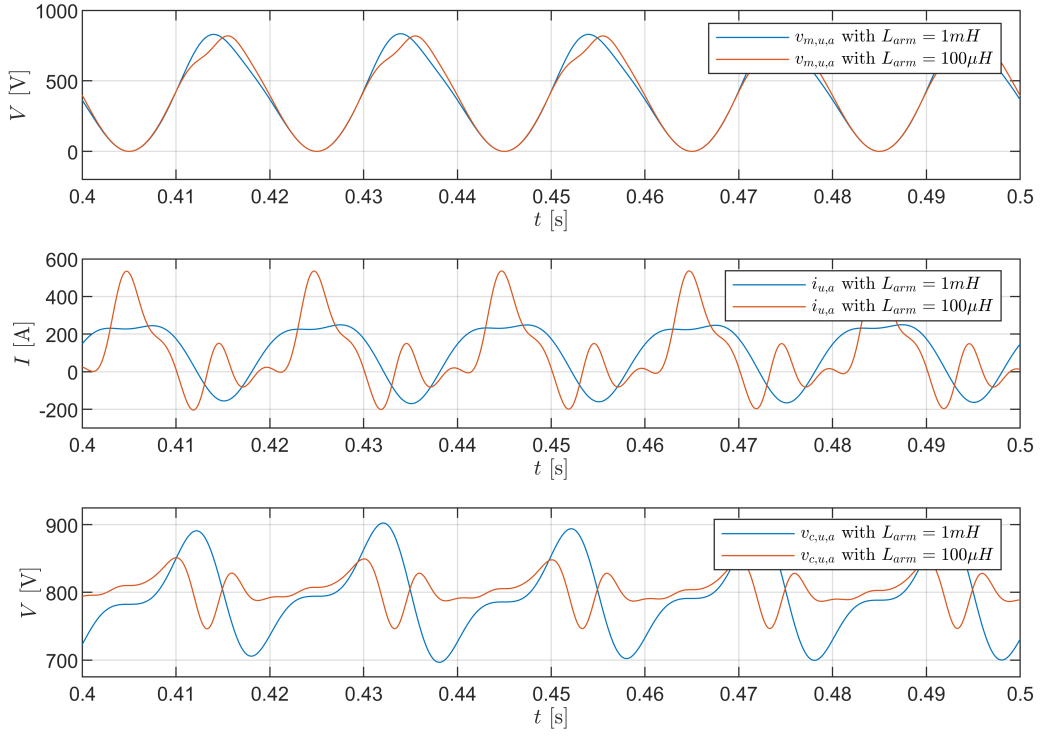


Figure 3.9: $\Delta\Sigma dq0$ frame outputs simulation results

3.3 Summary

The “average” model have an equivalent capacitor added to the system, introducing the modules charging and discharging dynamics. From this, we can directly relate the presence of a circulating current of 2ω with the capacitor dynamics, since a 2ω component was not present in the first model.

Chapter 4

MMC switching model (detailed)

4.1 Model on simulink

The switching model simulation has the same structure as the average model shown in [Figure 3.2](#), but the MMC model block is modified to represent the MMC SM switching as shown in [Figure 4.1](#).

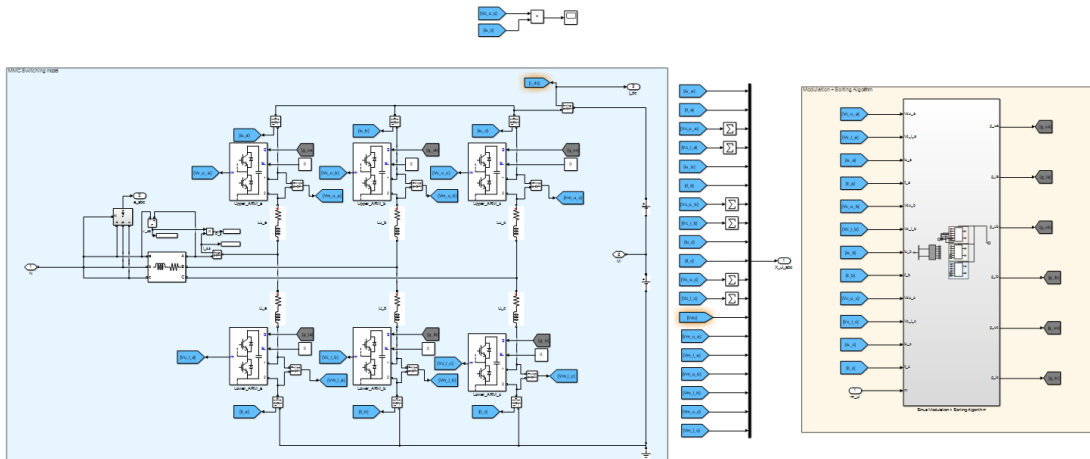


Figure 4.1: MMC switching model schematic

For this, the "Half-bridge MMC" block is used to model the MMC arms. It receives an input g that is the firing pulses for the two switches of each module, k_1 and k_2 . Its parameters in [Figure 4.2](#) are the number of SM in each arm and the SM capacitor calculated from the equivalent capacitor. The capacitors are said to be charged to their ideal mean voltage $\frac{V_{dc}}{N}$ for the simulation initial conditions.

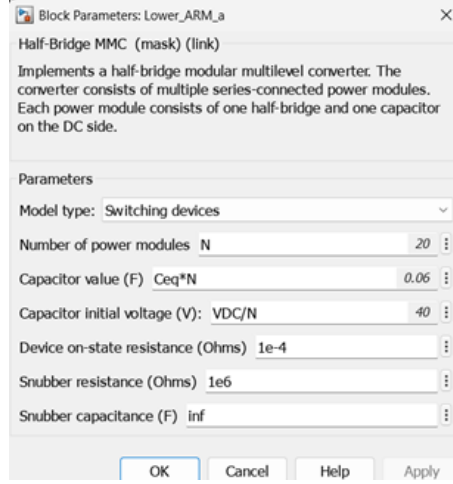


Figure 4.2: Half-bridge MMC block parameters

On the right, we can see that one extra block was added to the MMC model. Its structure is shown in [Figure 4.3](#). It corresponds to the PWM strategy used (SPWM) and the Voltage Balancing Algorithm (VBA), allowing to generate the firing pulses for each MMC's SM.

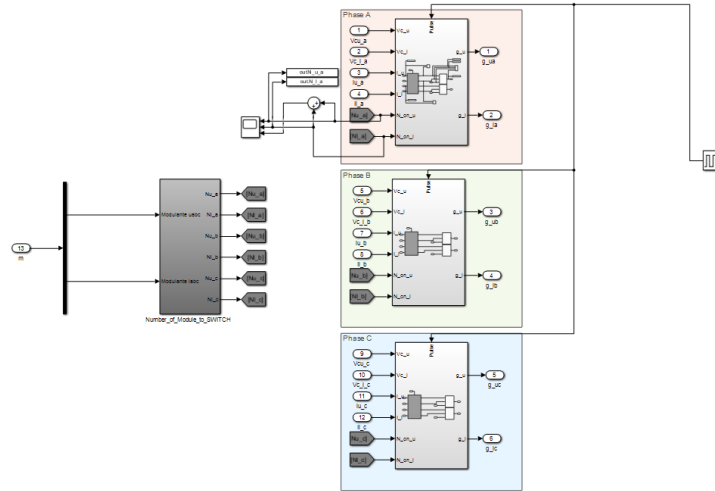


Figure 4.3: Modulation + VBA block

4.1.1 SPWM

As the converter is modeled with IGBT/diode pairs on the modules, a multilevel PWM generator is needed. The PWM produces the firing pulses (0/1 signals), which traditionally trigger switching in the converter.

In the case of the MMC, the PWM strategy is used to define the number of modules to be connected in each arm. [Figure 4.4](#) describes the different PWM stages. First, the modulating signal for the upper and lower arms of each phase are adapted before being applied to the PWM block.

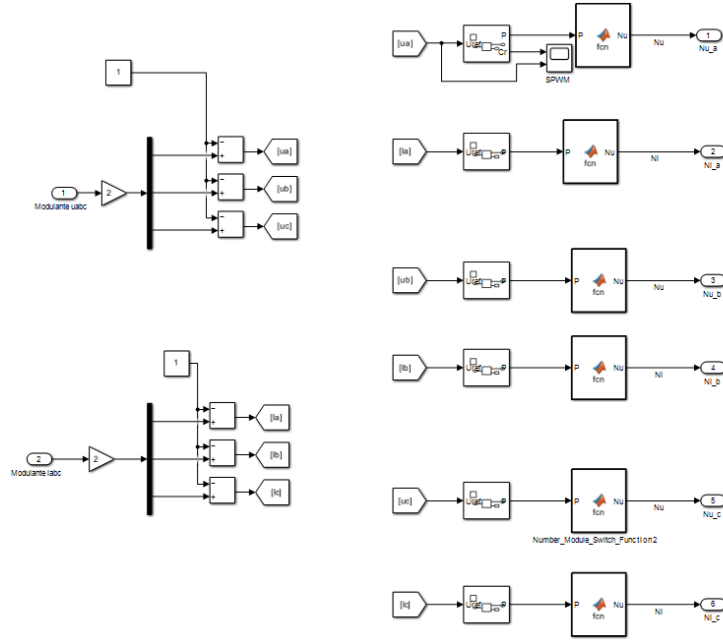


Figure 4.4: PWM block in simulink

The PWM technique used is SPWM, that is a multiple carriers PWM strategy. [Figure 4.5](#) shows that, in the SPWM strategy, N carriers phase-shifted of $\frac{360^\circ}{N}$ are used and compared to the modulating signals ($u_{u,p}$ and $u_{l,p}$), which defines the number of SM to be connected to the upper and lower arms instantaneously.

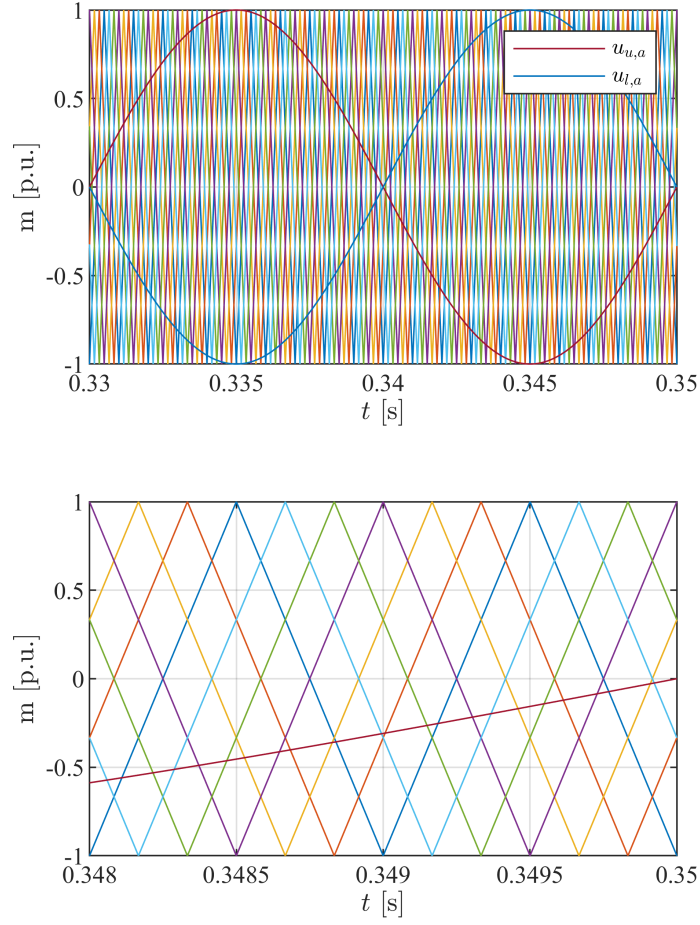


Figure 4.5: SPWM strategy functioning

In [Figure 4.6](#) obtained through simulations using $N = 6$, we see the number of SM to be inserted in the upper ($N_{u,p}$) and lower arm ($N_{l,p}$) generated by the SPWM. It is always contained between 0 and N and the sum of the $N_{u,p}$ and $N_{l,p}$ is equal to N , the total number of SM in each arm. This happens to maintain the V_{dc} input voltage level.

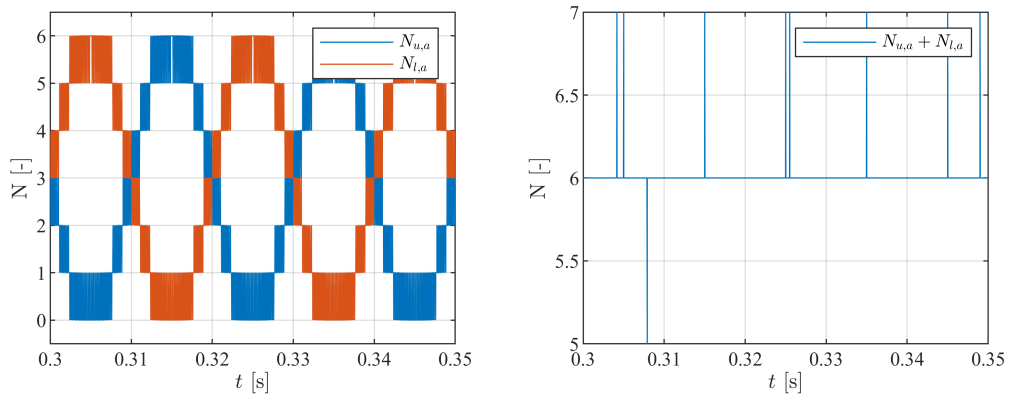


Figure 4.6: Simulated number of inserted SM

4.1.2 VBA sorting algorithm

After the number of SM is defined, the VBA (Voltage Balancing Algorithm) choses which SMs are going to be inserted within the upper and lower arms. Figure 4.7 shows how the VBA is implemented in the Simulink simulation.

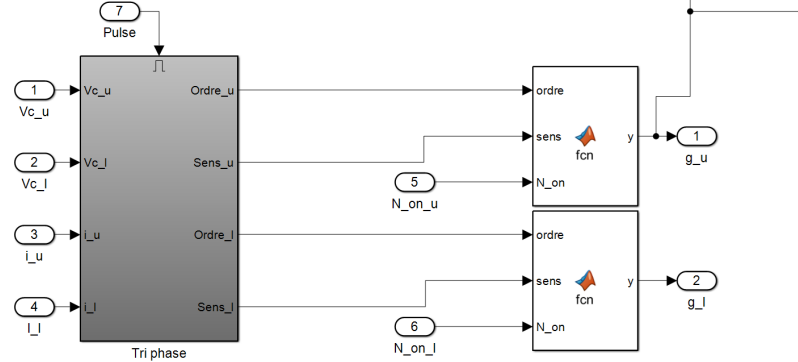


Figure 4.7: VBA block in simulink

This selection takes as inputs the SM capacitor voltages, the measured current passing through the arms, the number of SM to be inserted from the SPWM and a clock pulse signal to synchronize the switching. Its objective is to select the capacitors in a way that the capacitors of each arm have their voltage balanced in practically the same mean level. The same VBA algorithm is used for all phases.

The VBA is composed of 2 blocks. In the first logical block on the left, it is defined the order of the SM sorted according to the capacitors voltages and an indicator the sign of arm current is positive (1) or negative (0). If the arm current is positive, it means that the inserted capacitors will be charged. Similarly, if the arm current is negative, the inserted capacitors will be discharged.

In the second block on the right, the real VBA algorithm is implemented. It takes the order of the SM with respect to the capacitors voltages and the indicator of the arm current sign as an input as well as the number of SM N_{on} to insert into the arm. It first evaluates if the current sign is positive or negative to verify if the connected modules are going to be charged or discharged. If the current is positive, the N_{on} SM with smaller capacitor voltage are inserted to the arm. Similarly, if the current is negative, the N_{on} SM with highest capacitor voltage are inserted to the arm.

The algorithm logic can be verified in Figure 4.8. When N_{on} changes from 1 to 2, we see that the arm current is positive. This means that the 2 SMs with lowest voltages will be inserted. According to the SMs order, the SMs with lowest voltages are the 6th and the 2th.

By observing the K_1 pulses, we can deduce which SMs are inserted in the arm. If $K_1 = 1$ and $K_2 = 0$, the SM is inserted, otherwise, if $K_1 = 0$ and $K_2 = 1$, the SM is bypassed. Carrying this analysis to Figure 4.8, the 6th is already inserted when N_{on} changes, therefore only the 2th SM is inserted into the arm (its K_1 changes to 1).

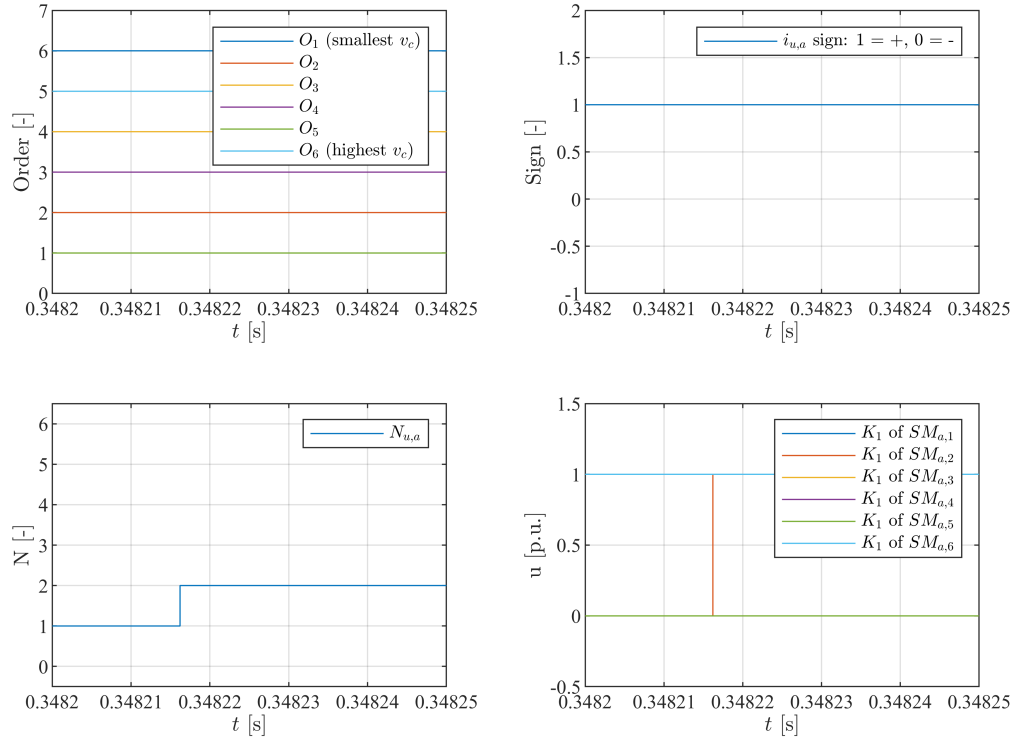


Figure 4.8: Test of VBA logic

4.2 Simulations

4.2.1 Results in ul, abc frame

From simulation results of Figure 4.9, we can clearly see that the voltage on the upper and lower arms have a ladder shape.

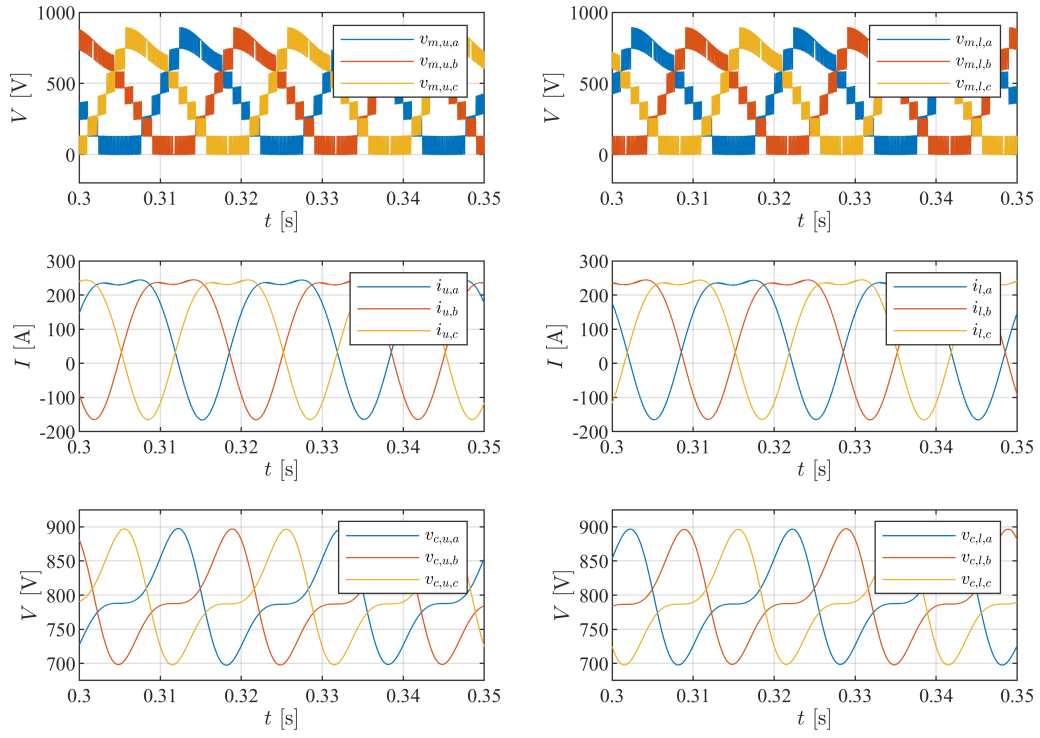


Figure 4.9: *ulabc* frame outputs for the SW model simulation

In Figure 4.10, the output voltage for phase a, $v_{s,a}$, have a nice sinusoidal shape by using only L_{arm} reduced filtering.

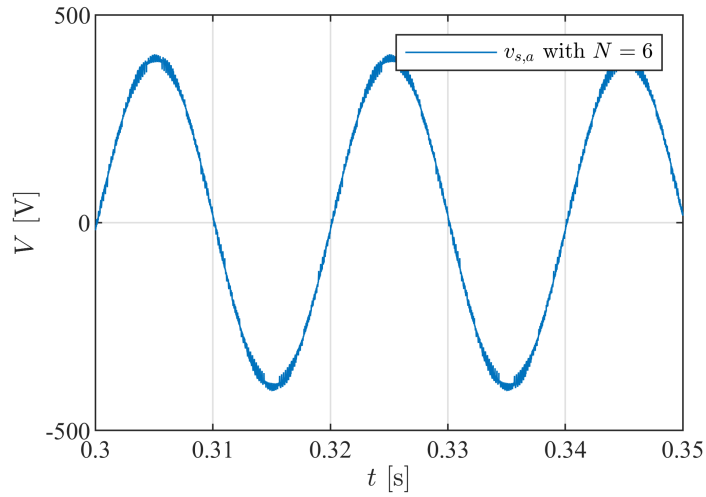


Figure 4.10: Voltage output on the load for the SW model simulation

4.2.2 Modifying N

Simulating the same system with a higher number of SM, $N = 20$, Figure 4.11 shows the MMC main outputs for only the upper phase a and Figure 4.12 shows the phase a voltage

output. By increasing N , we obtain a shape closer to a sinus on the arm voltages when compared to $N = 6$. Consequently, the converter voltage output have fewer harmonics.

For both $N = 6$ and $N = 20$ the AC voltage output is close enough to a sinus to exclude the need of filters on the MMC AC output.

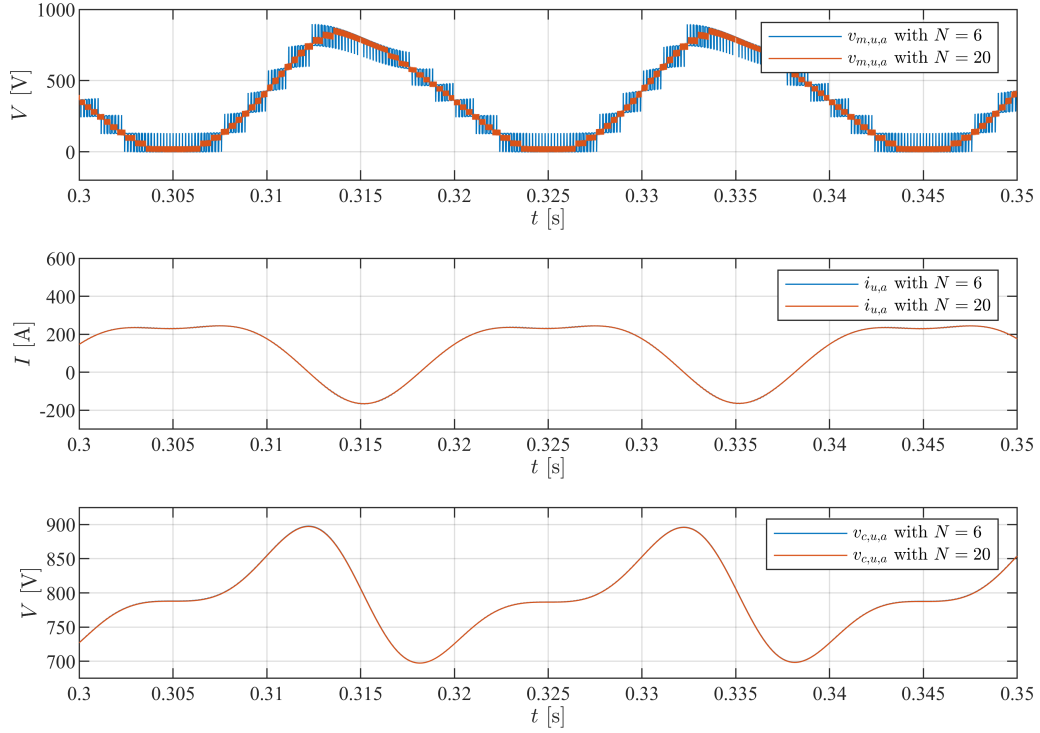


Figure 4.11: *ulabc* frame outputs for the SW model simulation with $N = 20$

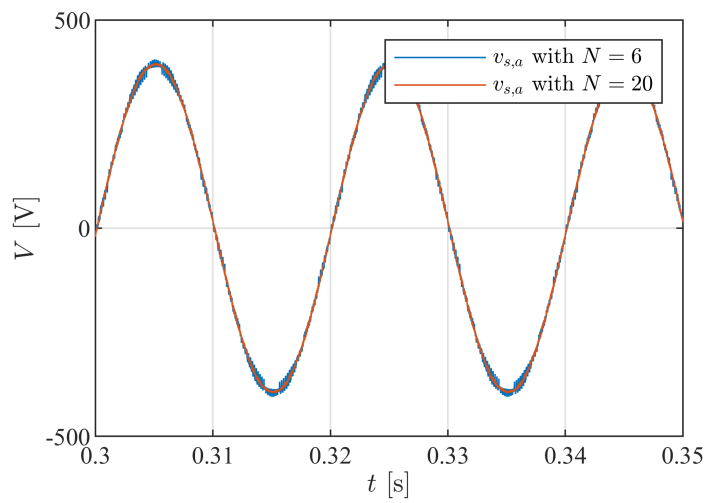


Figure 4.12: Voltage output on the load for the SW model simulation with $N = 20$

4.3 Summary

The switching model is different from the average model because it includes the power switches on its model. Because of that, it needs to represent the low-level control strategies composed of a PWM modulation + Voltage Balancing Algorithm (VBA) allowing to chose the SMs state at each time.

From simulation results, we observed that the voltages produced on the MMC's arms are in a ladder form that becomes closer to a sinus by increasing the number of SMs. Also, the SW model of the MMC presents fewer harmonics if compared to other topologies, e.g. the 2-level converter, such that there is no need for filters on the converter output.

Chapter 5

Models comparison

Figure 5.1 and Figure 5.2 shows a comparison of the models in terms of the simulation results.

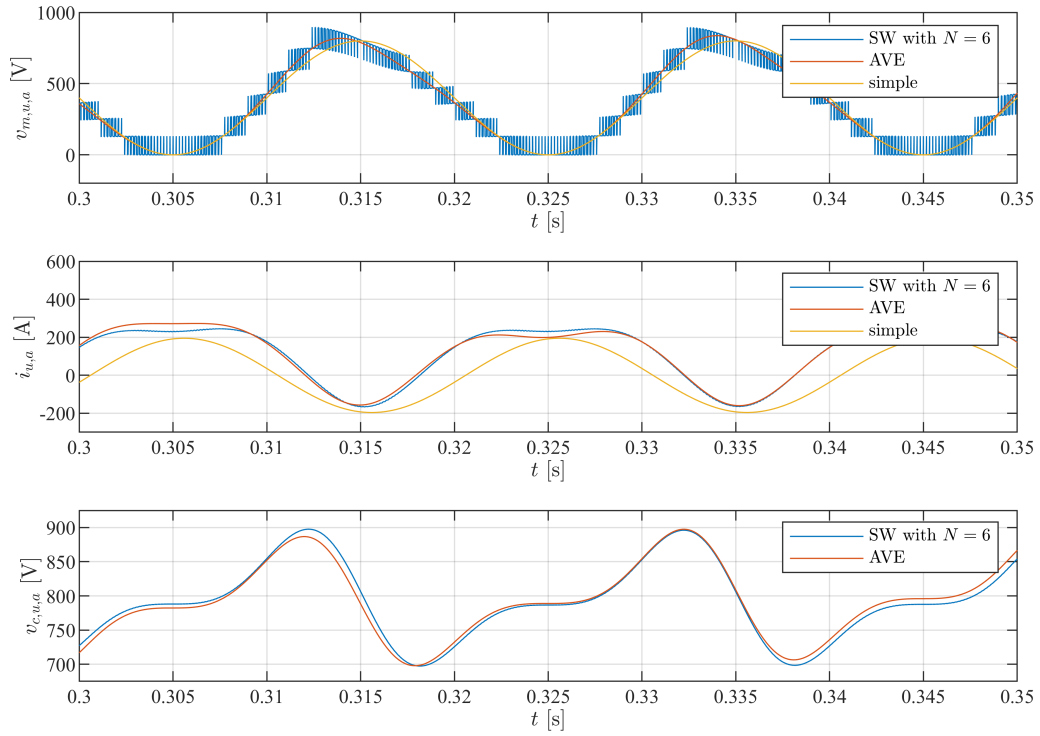


Figure 5.1: Multiple output quantities for all models

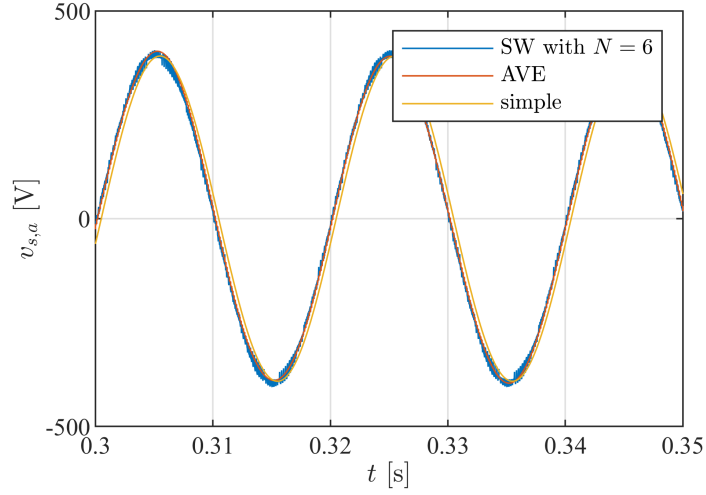


Figure 5.2: Voltage output on the load for all models

The simple model is overly idealized and does not adequately represent the complexity of MMC SMs and, unlike the other two models, it does not model the capacitors voltage.

Meanwhile, the average model provides an accurate representation of MMC operation when a sufficient number of SMs are used, as its results closely align with those of the SW model for $N = 20$.

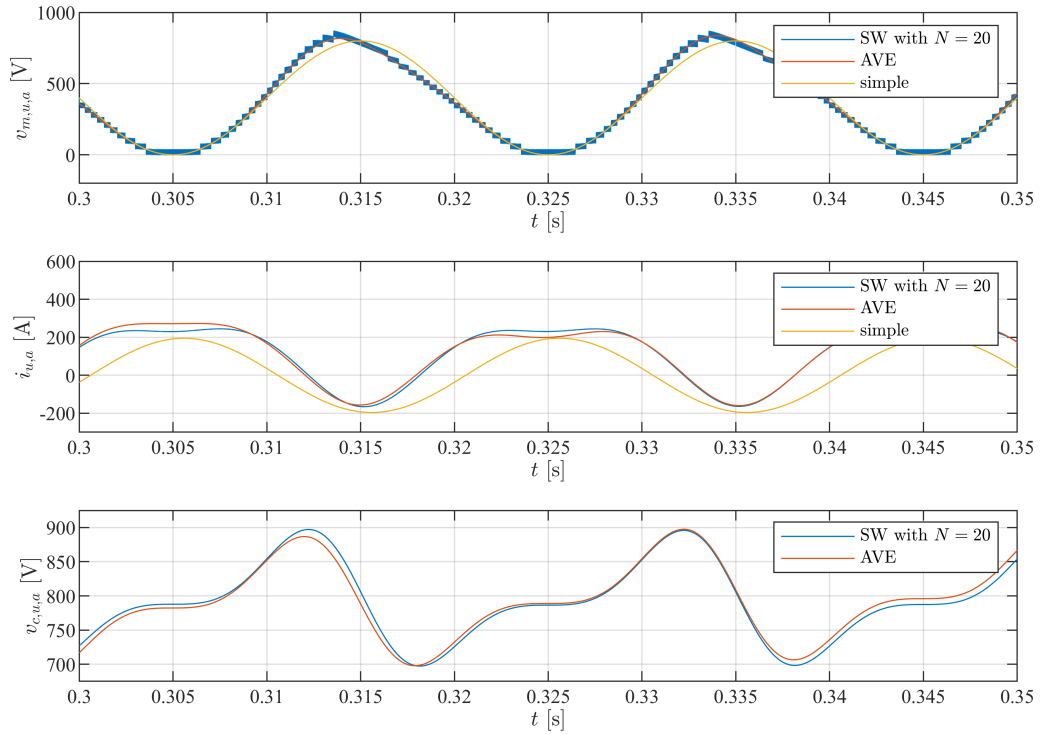


Figure 5.3: Multiple output quantities for all models

Bibliography

- [1] R. Coelho Medeiros, "Cryo-MMC - a Modular Multilevel Converter with Superconducting Coupled Arm Coils," PhD thesis, Université Paris-Saclay, Paris-Saclay, France, 2022.
- [2] B. Džonlaga, "Contribution to the sizing of the modular multilevel converter," PhD thesis, Université Paris-Saclay, Paris-Saclay, France, 2019.
- [3] G. Bergna, J. A. Suul and S. D'Arco "State-space modelling of modular multilevel converters for constant variables in steady-state" *2016 IEEE 17th Workshop on Control and Modeling for Power Electronics (COMPEL)*, 2016.